Practice workbook · Rectangle and shape problems with inequalities

Word problems where lengths in a diagram give expressions for perimeter or area, which are then used to form and solve equations or inequalities.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-04

1. Question 2 · part whole · 6 marks

2024 NOV · Paper 02

Worked-example retry · Attempt part whole only.

2 The length of rectangle R is 2 cm greater than its width.

The area of R is greater than $8\,cm^2$ and the perimeter of R is less than 30 cm.

Given that the width of R is w cm,

find the set of possible values of w
Give your answer in the form $a < w < b$ where a and b are rational numbers.

(6)

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P 7 3 5 8 7 A 0 4 3 2

1. Question 2 · part whole · 6 marks

2024 NOV · Paper 02

Worked-example retry · Attempt part whole only.

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Question 2 continued

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(Total for Question 2 is 6 marks)

1. Question 2 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 4 · part a · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

4



Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a rectangle with width x metres and length $(x + 4)$ metres.

The perimeter of the rectangle is P metres and the area of the rectangle is $A\,m^2$

(a) Find, in terms of x, an expression for

(i) P (ii) A

(2)

The perimeter of the rectangle has to be less than 30 metres.
The area of the rectangle has to be greater than $12\,m^2$

(b) Find the set of possible values for x
Give your answer in the form $a < x < b$

(5)

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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3. Question 4 · part b · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

4



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3. Question 4 · part b · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 7 marks)

3. Question 4 · part b · 5 marks · Working and answer

Working and answer

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4. Question 5 · part whole · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

5



Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a rectangle with width x cm and length $(2x+3)$ cm.

The perimeter of the rectangle is P cm and the area of the rectangle is A cm^2

$$P > 10 \quad \text{and} \quad A < 35$$

Find the set of possible values for x
Give your answer in the form $a < x < b$ where a and b are rational numbers.
Show clear algebraic working.

(7)

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4. Question 5 · part whole · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 7 marks)

4. Question 5 · part whole · 7 marks · Working and answer

Working and answer

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5. Question 2 · part a · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a only.

2

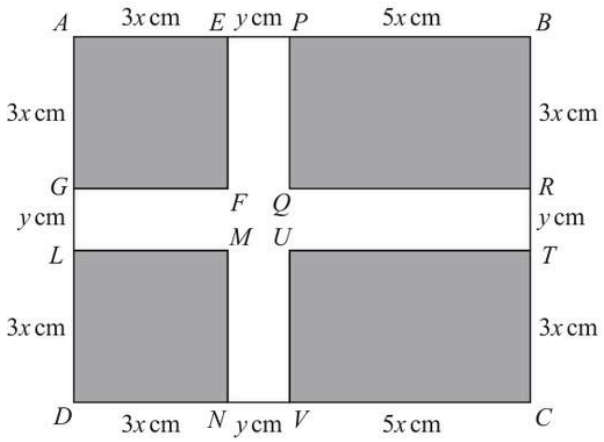


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the design for a flag consisting of a white cross on a grey background.

AEFG and *DLMN* are squares with sides of length $3x$ cm.
BPQR and *CTUV* are rectangles with sides of length $5x$ cm and $3x$ cm.
The width of the cross is y cm.

The total area of the flag is H cm^2

(a) Write down an expression, in terms of x and y, for H.

(1)

Given that the area of the cross is K cm^2

(b) show that $K = 14xy + y^2$

(3)

The total area of the flag is to be 3432 cm^2 and the area of the cross is to be 1080 cm^2

(c) Find the value of x and the value of y

(5)

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P 7 1 6 6 1 A 0 4 3 2

5. Question 2 · part a · 1 mark · Working and answer

Working and answer

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6. Question 2 · part b · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

2

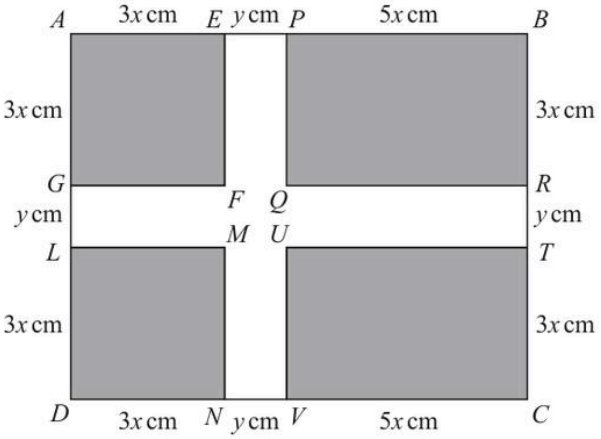


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6. Question 2 · part b · 3 marks · Working and answer

Working and answer

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